Excel Engineering College, Autonomous

Department of ECE

Title of the Project: Solar Panel Defect Detection Using Thermal and Visual Images Based on MATLAB Image Processing

Domain: Digital Image Processing

Name of the Supervisor with Designation:
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Presented by

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Problem Statement

- Manual inspection of solar panels is slow, labour-intensive, and can fail to identify hidden internal faults.
- Defects such as dust, cracks, hotspots, and delamination reduce the efficiency and overall performance of panels.
- Existing methods analyse thermal or visual images separately, so each misses defects the other modality would catch.
- A low-cost, automated way to combine thermal and visual data is needed to detect defects early and protect energy yield.

Objectives

- To detect defects from thermal and visual images.
- Classify cracks, hotspots, dust, and shading using segmentation.
- Fuse thermal and visual data for higher detection accuracy.
- Generate automated defect reports for maintenance prioritisation.
- Develop the system in MATLAB as a lightweight, transparent alternative to deep learning.

Literature Review

S.No	Paper	Author	Technique	Key Finding	Limitations
1	Thermal Imaging for PV Fault Detection	R. Kumar (2025)	Thermal Image Segmentation	Detects hotspots caused by cell damage or shading	Sensitive to ambient temperature variation
2	Visual Crack Detection in Solar Panels	S. Meena (2024)	Edge Detection & Morphology	Accurately isolates surface cracks and micro-cracks	Struggles under poor lighting
3	Dust Detection using Image Processing	K. Ravi (2023)	Colour & Texture Analysis	Identifies dust accumulation reducing output	Needs periodic recalibration for soil type
4	Deep Learning based PV Defect Classification	A. Sharma (2025)	CNN Classification	High accuracy across multiple defect categories	Needs large labelled dataset
5	Thermal-Visual Image Fusion for Fault Diagnosis	P. Nandhini (2024)	Image Fusion Techniques	Improves accuracy over single-mode imaging	Higher computational cost for real-time use

Research / Design Gap

- Existing methods use thermal, visual, or deep-learning approaches separately.
- Very few studies combine thermal and visual imaging within one MATLAB pipeline.
- CNN methods are accurate but need large labelled datasets and heavy computation.
- Very few systems generate a structured, automated maintenance report.
- Gap addressed: a lightweight MATLAB pipeline that fuses thermal and visual images and outputs an automated defect report.

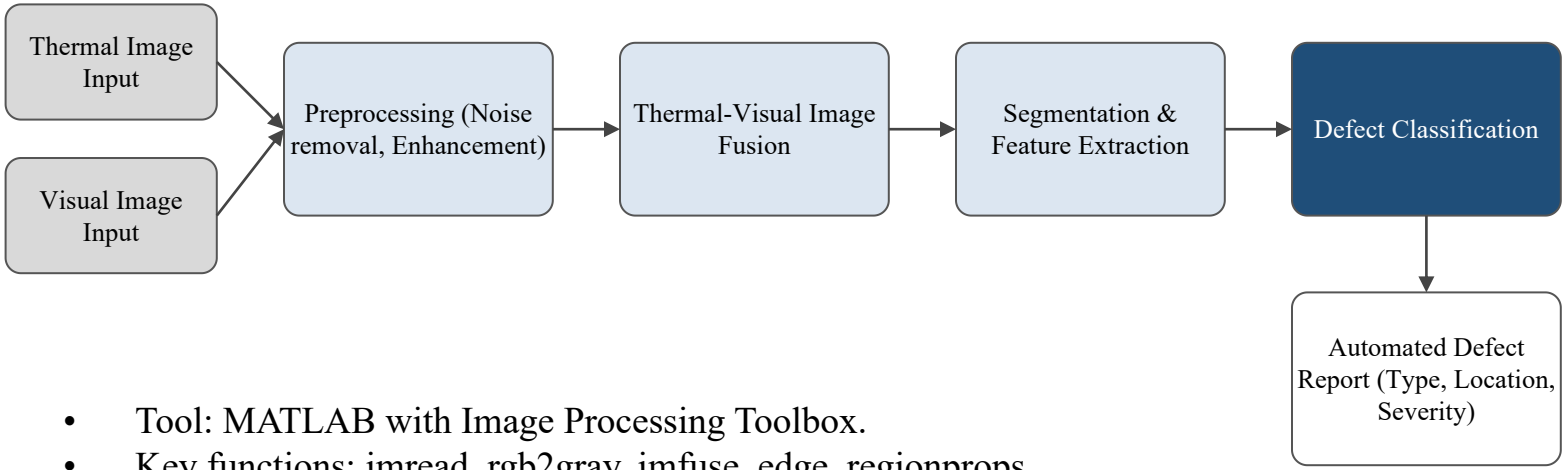
Existing System

- Manual visual inspection by technicians — labour-intensive, subjective, and blind to internal or thermal faults.
- Single-mode imaging — only a thermal (IR) camera or only a visual camera is used, so each modality misses what the other reveals.
- Deep-learning / CNN based systems — accurate, but need large labelled datasets and heavy computing resources.
- Limitation: no thermal-visual fusion, high cost, and poor scalability for small and mid-size installations.

Proposed System

- Pre-process thermal and visual images of the panel — noise removal and enhancement.
- Fuse the thermal and visual images to combine their complementary information.
- Segment and extract features to isolate candidate defect regions.
- Classify each region as crack, hotspot, dust, or delamination.
- Generate an automated report with defect type, location, and severity.
- Advantages: rule-based and transparent, needs no large training dataset, lightweight and cost-effective.

Block Diagram / Tools Used



- Tool: MATLAB with Image Processing Toolbox.
- Key functions: imread, rgb2gray, imfuse, edge, regionprops.
- Input: thermal (IR) and visual (RGB) panel images.
- Output: defect-classified image and automated report.

Methodology / Algorithm

1. Image Acquisition — collect paired thermal (IR) and visual (RGB) images of each panel.
2. Preprocessing — noise removal, contrast enhancement, and registration of the image pair.
3. Image Fusion — combine thermal and visual images using imfuse to highlight defect areas.
4. Segmentation — rgb2gray, edge detection, and thresholding isolate candidate defect regions.
5. Feature Extraction — regionprops extracts shape, area, and intensity features per region.
6. Defect Classification — rule-based classification into crack, hotspot, dust, or delamination.
7. Report Generation — compile defect type, location, and severity into an automated report.

Hardware & Software Requirements

Hardware

- Computer / laptop with adequate processor and RAM

Software

- MATLAB with Image Processing Toolbox
- Windows operating system
- Key functions: imread, rgb2gray, imfuse, edge, regionprops
- Sample dataset of paired thermal-visual images

Implementation Progress

Task	Status
Literature survey & problem identification	Completed
System design & block diagram finalisation	Completed
MATLAB preprocessing module (noise removal, enhancement)	Completed
Thermal-visual image fusion module	In Progress
Segmentation & feature extraction	Completed
Defect classification logic	Upcoming
Automated report generation	Upcoming

Preliminary Results

- Sample thermal-visual image pairs preprocessed successfully — noise reduced, contrast enhanced.
- Initial fusion output clearly shows combined thermal and visual defect regions.
- Edge detection isolates surface cracks correctly on test images.
- Hotspot regions identified through thermal intensity thresholding.

Sample MATLAB Code (Preprocessing to Feature Extraction):

```
IT = imread('thermal_panel.jpg');
IV = imread('visual_panel.jpg');
Ig = rgb2gray(IV);
If = imfuse(IT, Ig, 'blend');
Ie = edge(rgb2gray(If), 'Canny');
stats = regionprops(Ie, 'Area', 'BoundingBox');
```

Challenges & Limitations

- Image Quality: Low-resolution or unclear images may affect defect detection accuracy.
- Lighting Variations: Changes in sunlight, shadows, and brightness can affect visual image processing.
- Thermal Image Dependency: Proper temperature differences are required to identify hotspots accurately.
- Segmentation Errors: Incorrect thresholding or segmentation may cause some defect regions to be missed.
- Limited Dataset: A small dataset may reduce the accuracy and reliability of the defect detection system.

Work Completed vs. Work Remaining

Completed

- Literature survey & gap analysis
- System design — block diagram & methodology
- Preprocessing module development
- Initial thermal-visual fusion testing
- Finish segmentation & feature extraction

Remaining

- Implement defect classification logic
- Build automated report generation module
- Full system testing, validation & final documentation

Reference

[1] M. Waqar Akram et al., “Improved Outdoor Thermography and Processing of Infrared Images for Defect Detection in PV Modules,” Solar Energy, 2019.

[2] Hale Bakr et al., “Automatic Detection of Deteriorated Photovoltaic Modules Using IRT Images and Deep Learning,” Eng. Applications of AI, 2023.

[3] M. Waqar Akram et al., “Defect Detection in Photovoltaic Modules Based on Image-to-Image Generation and Deep Learning,” 2025.

[4] Keony Jimenez, Juan B. Cano & Esteban Velilla, “A Lightweight Deep Learning Model for Fault Detection of PV Modules Using Thermal Images,” Solar Energy, 2026.